

Day 9 - AM - List Deep Dive

Part A :- Concept Application (Student Management System)

File Link:-

https://github.com/NiranjjanHebli/week2/blob/main/day9/lists/student_system/student_system.py

Output Screenshots:-

```
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 student_system.py

+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+

1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 1
Name      : Mihir Gor
Subject (Math / Physics / Chemistry): Math
Marks     : 90
Added: Mihir Gor | Math | 90
```

```
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 student_system.py
```

```
7 Exit and Save
Enter choice (1-7): 1
Name : Mihir Gor
Subject (Math / Physics / Chemistry): Math
Marks : 90
Added: Mihir Gor | Math | 90
```

```
+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+
```

```
1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 6
```

#	Name	Subject	Marks
1	Aman	Math	88
2	Priya	Physics	91
3	Rahul	Math	76
4	Sneha	Chemistry	84
5	Arjun	Physics	67
6	Meera	Math	95
7	Karan	Chemistry	78
8	Divya	Physics	82
9	Rohan	Chemistry	90
10	Tanvi	Math	73
11	Mihir Gor	Math	90

Latest record: Mihir Gor | Math | 90

```
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 student_system.py
```

1	Aman	Math	88
2	Priya	Physics	91
3	Rahul	Math	76
4	Sneha	Chemistry	84
5	Arjun	Physics	67
6	Meera	Math	95
7	Karan	Chemistry	78
8	Divya	Physics	82
9	Rohan	Chemistry	90
10	Tanvi	Math	73
11	Mihir Gor	Math	90

Latest record: Mihir Gor | Math | 90

```
+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+
```

```
1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 2
Enter subject (Math / Physics / Chemistry): Math
```

Top 3 Toppers - Math

Rank	Name	Marks
1	Meera	95
2	Mihir Gor	90
3	Aman	88

```
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 student_system.py
```

```
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 3
Enter subject (Math / Physics / Chemistry): Physics
```

Class Average - Physics: 85.0

```
+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+
```

```
1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 4
```

Overall Average: 84.5
Students scoring above average:

Name	Subject	Marks
Aman	Math	88
Priya	Physics	91
Meera	Math	95
Rohan	Chemistry	90
Mihir Gor	Math	90
Niranjan Hebli	Physics	100

```
+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+
```

```
1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 5
Enter student name to remove: Niranjan Hebli
Removed 1 record(s) for 'Niranjan Hebli'.
```

```
+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+
```

```
1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 6
```

#	Name	Subject	Marks
1	Aman	Math	88
2	Priya	Physics	91
3	Rahul	Math	76
4	Sneha	Chemistry	84
5	Arjun	Physics	67
6	Meera	Math	95
7	Karan	Chemistry	78
8	Divya	Physics	82
9	Rohan	Chemistry	90
10	Tanvi	Math	73
11	Mihir Gor	Math	90

Latest record: Mihir Gor | Math | 90

```
#      Name      Subject      Marks
-----
1      Aman      Math      88
2      Priya     Physics  91
3      Rahul     Math     76
4      Sneha     Chemistry 84
5      Arjun     Physics  67
6      Meera     Math     95
7      Karan     Chemistry 78
8      Divya     Physics  82
9      Rohan     Chemistry 90
10     Tanvi     Math     73
11     Mihir Gor  Math     90
Latest record: Mihir Gor | Math | 90

+-----+
|          STUDENT MANAGEMENT SYSTEM          |
+-----+

1 Add Student
2 Show Toppers (by Subject)
3 Show Class Average (by Subject)
4 Show Above-Average Students
5 Remove Student
6 Show All Records
7 Exit and Save
Enter choice (1-7): 7

Records saved to 'students.txt'.
```

Part B :- Stretch Problem (Matrix Operations)

File link:-

https://github.com/NiranjanHebli/week2/blob/main/day9/lists/matrix_ops.py

Output Screenshot:-

```

(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 matrix_ops.py
Matrix Operations - Test Cases

2x2 Matrices

A =
[[1, 2],
 [3, 4]]

B =
[[5, 6],
 [7, 8]]

matrix_add(A, B)      =

[[6, 8],
 [10, 12]]

matrix_transpose(A)   =

[[1, 3],
 [2, 4]]

matrix_multiply(A, B) =

[[19, 22],
 [43, 50]]

3x3 Matrices

C =
[[1, 2, 3],
 [4, 5, 6],
 [7, 8, 9]]

D =
[[9, 8, 7],
 [6, 5, 4],
 [3, 2, 1]]

matrix_add(C, D)      =

[[10, 10, 10],
 [10, 10, 10],
 [10, 10, 10]]

matrix_transpose(C)   =

[[1, 4, 7],
 [2, 5, 8],
 [3, 6, 9]]

matrix_multiply(C, D) =

[[30, 24, 18],
 [84, 69, 54],
 [138, 114, 90]]

Non-square (2x3) x (3x2)
E (2x3) = [[1, 2, 3], [4, 5, 6]]
F (3x2) = [[7, 8], [9, 10], [11, 12]]
E x F    = [[58, 64], [139, 154]]

Dimension Mismatch Handling
matrix_add mismatch      - Dimension mismatch for addition: (1x2) vs (1x3)
matrix_multiply mismatch - Dimension mismatch for multiplication: (1x2) x (1x2) - columns of A (2) must equal
rows of B (1)

```

Part C :- Interview Ready

Q1. Conceptual

Shallow copy creates a new object but references the same nested objects as the original, while deep copy recursively duplicates all levels for full independence.

Example with Nested Lists:-

```
import copy

original = [[1, 2], [3, 4]]
shallow = copy.copy(original)    # Shallow copy
deep = copy.deepcopy(original)   # Deep copy

shallow[0][0] = 99               # Modifies nested list
deep[0][0] = 88                  # Independent change

print(original) # [[99, 2], [3, 4]] - affected
print(shallow)  # [[99, 2], [3, 4]]
print(deep)     # [[88, 2], [3, 4]]
```

Why Shallow Copy Fails

Shallow copy only duplicates the top-level list, so inner lists remain shared references. Modifying a nested element (like `shallow[0][0]`) changes the original because both point to the same mutable object.

When Deep Copy Is Required

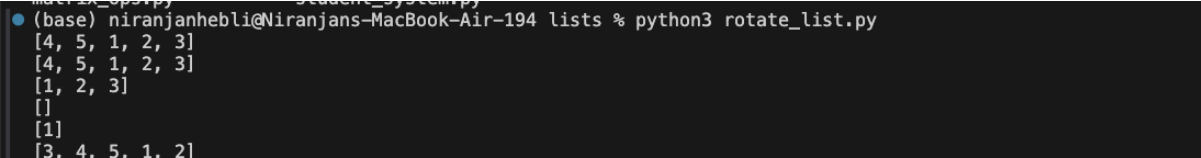
Use of deep copy is required for nested mutable structures (lists/dicts of lists/dicts) where independent modifications are needed, like data processing pipelines or when passing copies to functions that might alter contents.

Q2. Coding — List Rotation

File Link:-

https://github.com/NiranjanHebli/week2/blob/main/day9/lists/rotate_list.py

Output Screenshot:-



```
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 rotate_list.py
[4, 5, 1, 2, 3]
[4, 5, 1, 2, 3]
[1, 2, 3]
[]
[1]
[3, 4, 5, 1, 2]
```

Q3 Debug Problem

The bug causes [1, 3, 5] instead of [1, 3, 5, 7] because modifying a list during iteration skips elements.

Why the Bug Happens

The `for num in nums:` loop uses an internal index that increments regardless of removals. When `num=2` (index 1) is removed, elements shift left 3 moves to index 1 but the loop advances to index 2 (6), skipping 3. With `[2,4,6,8]`, it empties unexpectedly as each removal skips the next even leading to final output as `[4,8]` and not `[]`.

Correct Solution

Use list comprehension to filter odds in one pass:

Python Code:-

```
nums = [1, 2, 3, 4, 5, 6, 7, 8]
nums = [num for num in nums if num % 2 != 0]
print(nums) # [1, 3, 5, 7]
```

File Link:-

https://github.com/NiranjnHebli/week2/blob/main/day9/lists/debug_problem.py

Part D - AI Augmented - Task

Prompt:-

Write a Python function `find_pairs(nums, target)` that returns all unique pairs of numbers from the input list `nums` which sum exactly to `target`, using only list comprehensions.

Requirements:

- Input: `nums = [1, 2, 3, 4, 5, 7]`, `target = 9`
- Output: `[(4, 5), (2, 7)]` (pairs as tuples, order doesn't matter)
- Return empty list `[]` if no pairs found
- Handle duplicates and ensure each pair appears only once
- Use list comprehension only (no loops)

AI - Generated Code:-

```
def find_pairs(nums, target):  
    """Find all unique pairs that sum to target using list comprehension only."""  
    return [(nums[i], nums[j]) for i in range(len(nums))  
            for j in range(i+1, len(nums))  
            if nums[i] + nums[j] == target]
```

File

link:-https://github.com/NiranjanHebli/week2/blob/main/day9/lists/find_pairs/find_pairs.py

Duplicates Pairs?

Yes, the generated AI - Code gave output as duplicate pairs

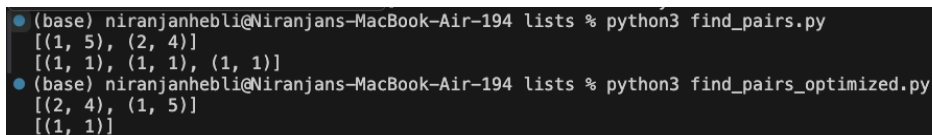
Incorrect Logic?

Yes, It had an incorrect logic since it didnt verify for duplicate pairs.

optimized code link:-

https://github.com/NiranjanHebli/week2/blob/main/day9/lists/find_pairs/find_pairs_optimized.py

Output Screenshots:-



```
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 find_pairs.py  
[(1, 5), (2, 4)]  
[(1, 1), (1, 1), (1, 1)]  
(base) niranjanhebli@Niranjans-MacBook-Air-194 lists % python3 find_pairs_optimized.py  
[(2, 4), (1, 5)]  
[(1, 1)]
```

Readme file containing analysis:-

https://github.com/NiranjanHebli/week2/blob/main/day9/lists/find_pairs/ai_sum_pairs.md